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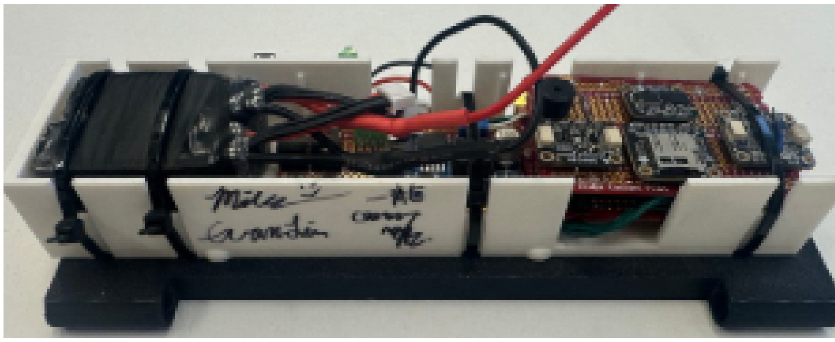
TRANSONIC ROCKET AVIONICS



More pictures:

What?

- 11 ft. level 3 rocket capable of **transonic speeds** and reuse, with a focus on safety and reliability
- Custom flight controller based on Raspberry Pi with **live telemetry**, onboard data logging, and dual parachute deployment

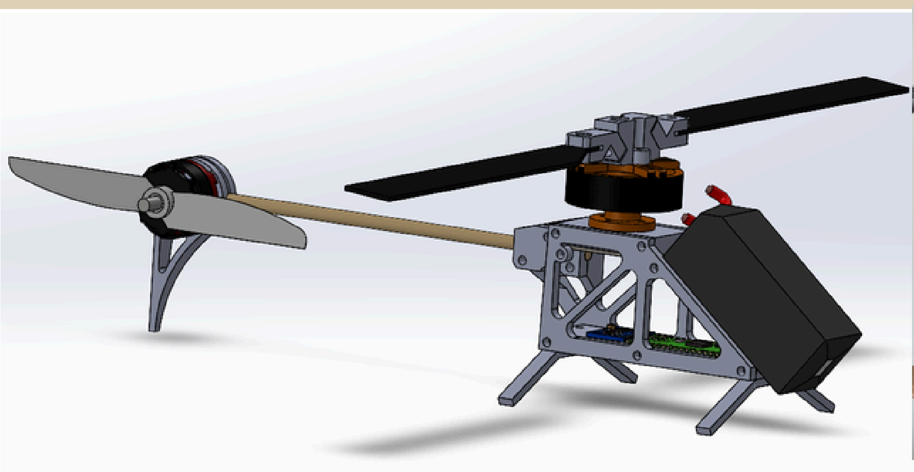
How?

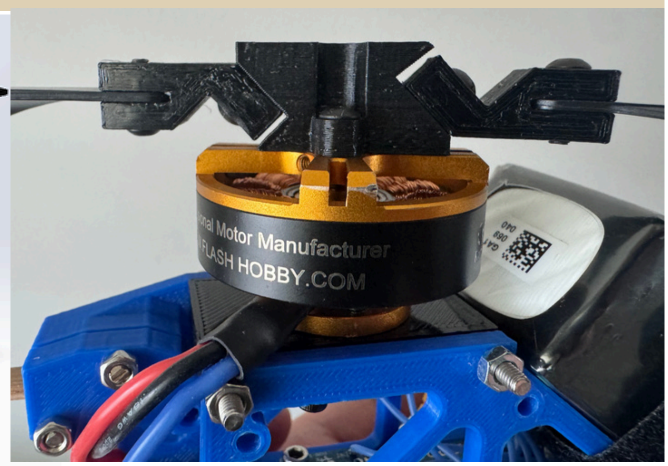
- Modeled avionics bay using NX Siemens
- Simulated aerodynamic properties using Open Rocket, **STAR-CCM+**, and **ANSYS**
- Manufactured fins, bulkheads, and coulplers using **composite layup** and **machining** methods
- Designed and prototyped a custom flight controller with **Altium Designer** and soldering methods

Results

- Passed technical inspection for L3 rockets at LDRS and launched successfully with an apogee of 8158 ft. AGL. Launch video:

CYCLIC THRUST MODULATED HELICOPTER





What?

- Designed and built a model helicopter with hinged blades and **specialized control loops** instead of a traditional swashplate mechanism
- Based on research papers from University of Pennsylvania:

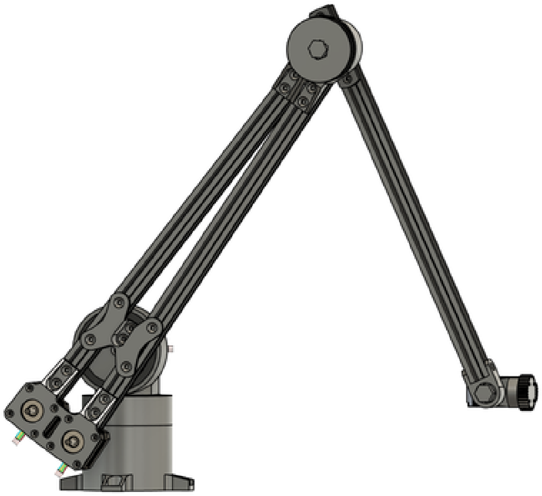
How?

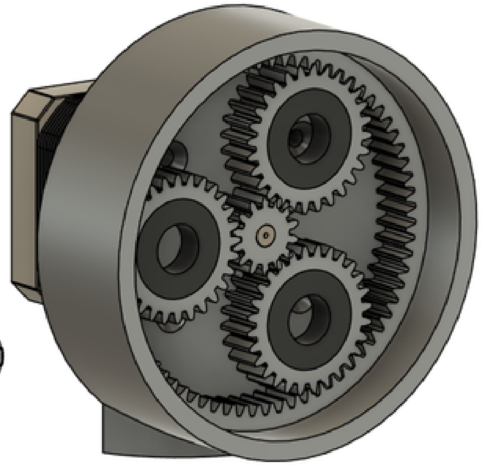
- Modeled using **SolidWorks**
- Simulated PID loops using **MATLAB Simulink**
- Wired a custom flight controller with inertial measurement unit, rotary encoder, RC receiver, battery, and motors

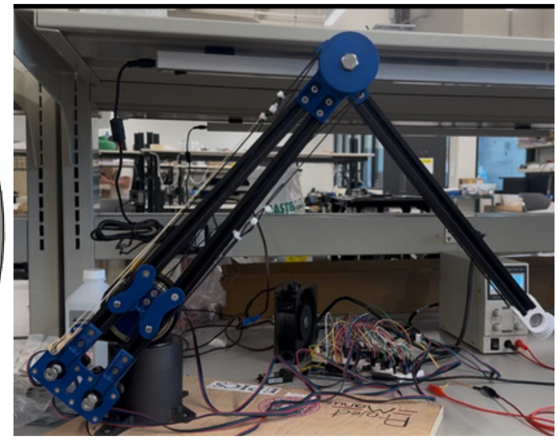
Results

- Full control over yaw, pitch, and roll while requiring **75% less moving parts** compared to swashplate rotors

ROBOTIC ARM WITH 5 DEGREES OF FREEDOM







What?

- Designed a 5 degree of freedom robotic arm based on **Arduino** and NEMA 17 **stepper motors**
- Optimized for ease of assembly and reproducibility with **3D printed** parts to encourage STEM education within high schools

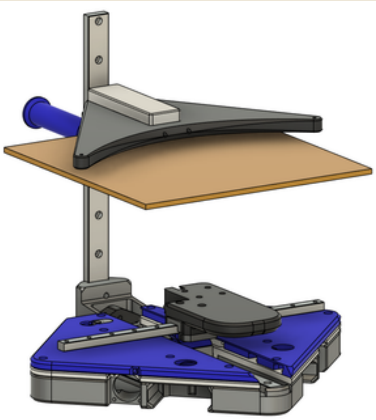
How?

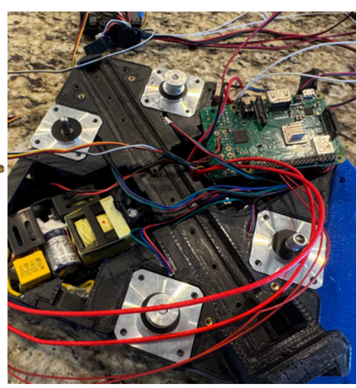
- Modeled all components using **Fusion 360**
- Incorporated interference and screw joints for simplicity
- Used geometry tools to generate gear tooth profiles suitable for 3D printing

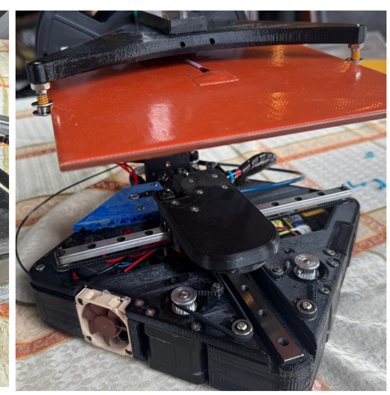
Results

- A final prototype has been assembled and plans have been made available for high schools
- **RC controlled** for accessibility
- Interchangeable end effectors
- **Belt-driven** to reduce moving weight
- Build guide, code, and wiring:

INVERTED COREXY 3D PRINTER







What?

- Designed a 3D printer with an **upside-down Core-XY** motion system to concentrate moving mass at the base
- Prioritized **space efficiency** with a high build volume to footprint ratio
- Developed a **robust and lightweight** toolend and bed holder

How?

- Used **SolidWorks** to fit all components around a 3D printed chassis
- Integrated fan ducts and wire channels for proper **heat dissipation**
- Implemented **DFMA principles** to reduce parts required and assembly cost
- Built using 3D printing and machining
- Installed and programmed a **Raspberry Pi** for printer control and file transfers

Results

- **400%** higher build area to footprint ratio compared to traditional printers (Prusa Mk4)
- Achieves print speeds of **220 mm/s**, reducing print time by **30%** on average
- Removable bed allows for quick turnaround time